Does It Rain More in Seattle Than in Berkeley?

Introduction

Seattle is widely known as a rainy city, but how much wetter is it than another West Coast city? This project compares precipitation in Seattle, Washington, and Berkeley, California. The two cities are located at similar longitudes, but Seattle is much farther north and has a different climate. The main question of this analysis is: **Does it rain more in Seattle than in Berkeley?**

Data and Methods

Daily precipitation data were obtained from the NOAA National Centers for Environmental Information using the Global Historical Climatology Network Daily dataset. The analysis covers January 1, 2018 through December 31, 2022.

Some daily precipitation observations were missing. Berkeley had 471 missing values, including several long gaps, so historical Berkeley precipitation data were used to estimate values based on similar times of the year. Seattle had 190 missing values, which were replaced using the average precipitation for the same day of the year across the available years.

I compared overall, monthly, and seasonal precipitation between the two cities. Welch's two-sample t-tests were used to compare monthly mean precipitation, while paired t-tests were used to compare seasonal precipitation between the cities across the five years.

Results

Seattle had substantially higher average daily precipitation over the five-year period. Seattle averaged approximately 0.113 inches per day, while Berkeley averaged approximately 0.053 inches per day.

The monthly comparison shows that the difference is especially noticeable during fall and winter. Berkeley becomes extremely dry during the summer, while Seattle continues to receive some precipitation.

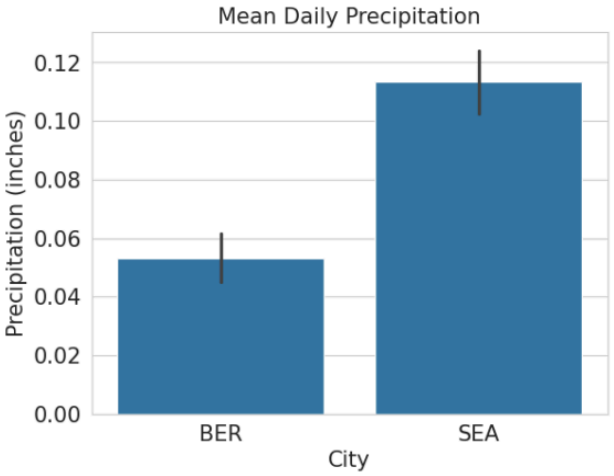


Figure 1. Mean Daily Precipitation: Seattle vs. Berkeley

Statistical tests of monthly precipitation found significant differences between Seattle and Berkeley in 10 of the 12 months. March (p-value = 0.279) and April (p-value = 0.059) were the only months where the differences were not statistically significant at the 0.05 level.

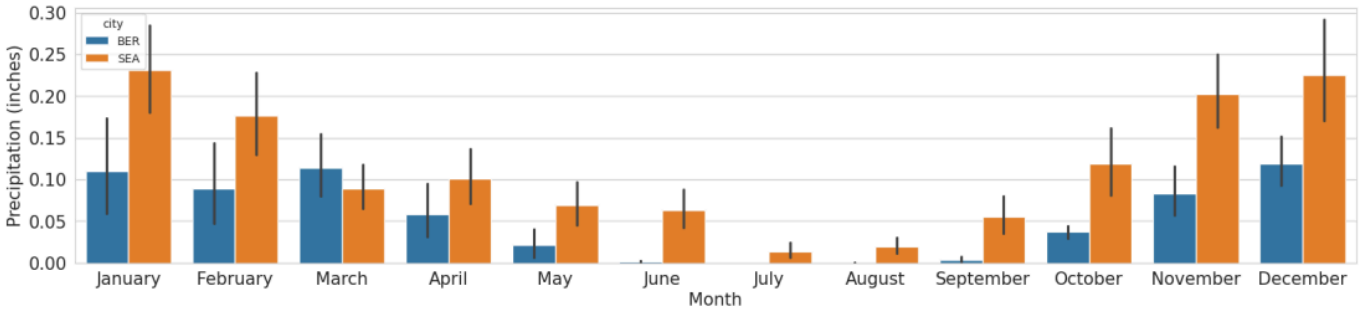


Figure 2. Mean Monthly Precipitation by City

The seasonal analysis showed a similar pattern. Seattle averaged 19.10 inches in winter compared with 9.62 inches in Berkeley, and 11.40 inches in fall compared with 3.77 inches in Berkeley. The largest relative difference occurred in summer: Seattle averaged 2.95 inches, while Berkeley averaged only 0.06 inches.

The seasonal differences were statistically significant in winter (p-value = 0.044), summer (p-value = 0.001), and fall (p-value = 0.011). Spring was the only season without a statistically significant difference (p-value = 0.369)

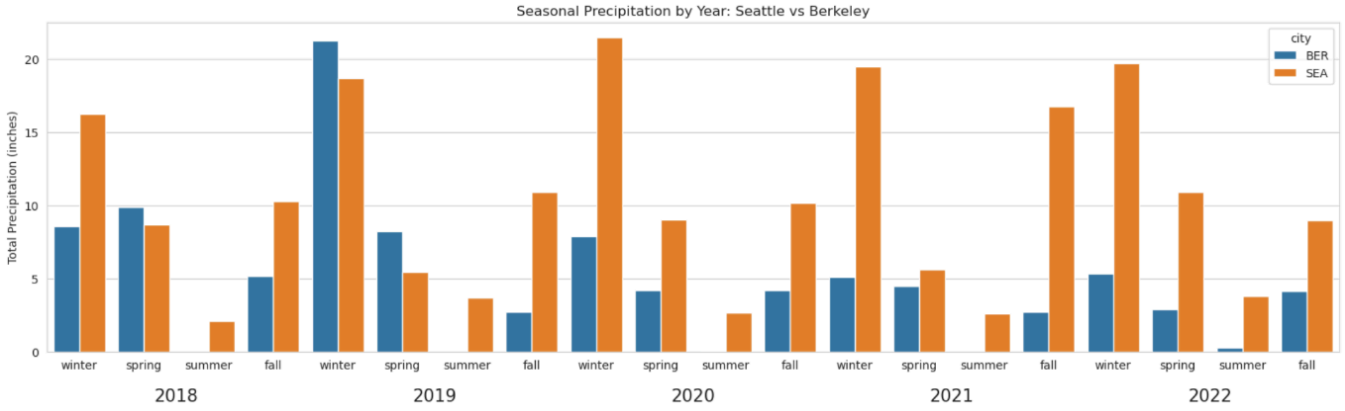


Figure 3. Seasonal Precipitation by Year: Seattle vs. Berkeley

Conclusion

The results support the conclusion that it rains more in Seattle than in Berkeley. Seattle had more than twice the average daily precipitation over the five-year study period, and the difference was visible throughout most months and seasons. The relative contrast was especially large during summer, when Berkeley received almost no precipitation while Seattle continued to receive some rain.

One limitation of this analysis is that some precipitation observations were missing and had to be estimated, particularly for Berkeley. In addition, the analysis covers only five years, so unusual weather during this period may influence the results. Future work could use a longer period of complete weather records or compare Seattle with additional West Coast cities to determine whether the same pattern remains consistent over time.